

Database Connection Guide

This document provides instructions for creating connection strings for PostgreSQL, MySQL, and MongoDB databases.

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1. PostgreSQL Connection String

Basic Format

```
postgresql://username:password@hostname:port/database_name
```

Examples:

1. Local Development:

```
postgresql://postgres:mysecretpassword@localhost:5432/mydb
```

2. Cloud Database (AWS RDS):

```
postgresql://dbuser:dbpassword@my-rds-instance.123456789012.us-east-1.rds.amazonaws.com:5432/production_db
```

3. With SSL:

```
postgresql://user:pass@host:5432/db?sslmode=require
```

Parameters:

Parameter	Description	Default
username	Database user	-
password	User password	-
hostname	Server address	localhost
port	Connection port	5432
database_name	Database name	-
sslmode	SSL requirement (disable/allow/prefer/require)	prefer

2. MySQL Connection String

Basic Format

```
mysql://username:password@hostname:port/database_name
```

Examples

1. Local Development (XAMPP):

```
mysql://root:@localhost:3306/mydb
```

2. Remote Server:

```
mysql://dbadmin:securepass@db.example.com:3306/production
```

3. With SSL Options:

```
mysql://user:pass@host:3306/db?ssl=true&sslmode=VERIFY_IDENTITY
```

Parameters:

Parameter	Description	Default
username	Database user	-
password	User password	-
hostname	Server address	localhost
port	Connection port	3306
database_name	Database name	-
ssl	Enable SSL	false

3. MongoDB Connection String

Basic Format

```
mongodb://username:password@hostname:port/database_name?options
```

Examples

1. Local Development:

```
mongodb://localhost:27017/mydb
```

2. Authenticated Connection:

```
mongodb://admin:secretpassword@localhost:27017/mydb?authSource=admin
```

3. Replica Set:

```
mongodb://user:pass@host1:27017,host2:27017,host3:27017/dbname?replicaSet=myReplicaSet
```

4. Atlas Connection:

```
mongodb+srv://cluster0.mongodb.net/mydb?retryWrites=true&w=majority
```

Parameters:

Parameter	Description	Default
username	Database user	-
password	User password	-
hostname	Server address	localhost
port	Connection port	27017
database_name	Database name	-
authSource	Authentication database	admin

replicaSet	Replica set name	-
ssl	Enable SSL	true (for Atlas)

Troubleshooting

Common Issues

- 1. Connection Refused**
 - Verify database server is running
 - Check firewall settings
 - Confirm correct port number
- 2. Authentication Failed**
 - Double-check username/password
 - Verify user has proper permissions
 - For MongoDB, check authSource parameter
- 3. SSL Errors**
 - For development, try `sslmode=disable` (PostgreSQL) or `ssl=false` (MySQL)
 - For production, ensure proper certificates are configured
- 4. Network Issues**
 - Verify hostname is reachable
 - Check for VPN requirements
 - Test with basic connection first, then add options

Testing Connections in Bash or CMD

PostgreSQL:

```
psql "postgresql://user:pass@host:port/db"
```

MySQL:

```
mysql -u user -p -h host -P port dbname
```

MongoDB:

```
mongosh "mongodb://user:pass@host:port/db"
```

This document covers the essential connection string formats for the three major database systems. Always keep your connection strings secure and never commit them directly to version control.

